

ANUSHKA PARIHAR

AI & Backend Engineer • 4th Year B.Tech CSE

+91-8827809748 | pariharanushka0206@gmail.com | GitHub | LinkedIn | Indore, Madhya Pradesh

SUMMARY

Computer Science undergraduate at SGSITS Indore with hands-on experience in distributed backend systems (Spring Boot, Kafka, microservices) and agentic AI (LangGraph, RAG, CrewAI, voice AI). Proven ability to ship end-to-end projects from JWT-secured APIs and event-driven architectures to LLM orchestration and real-time voice agents.

EDUCATION

Bachelor of Technology — Computer Science Engineering

Shri Govindram Seksaria Institute of Technology and Science (SGSITS), Indore

12th Grade — Higher Secondary (CBSE), Pioneer Convent School, Indore

CGPA: 7.1

2023 – 2027

80.2%

TECHNICAL SKILLS

Languages: Python, C++, Java

AI / ML: Agentic AI, CrewAI, LangGraph, OpenAI Realtime API, RAG Pipelines, LLM Orchestration, ChromaDB, FastAPI

Backend & Microservices: Spring Boot, Spring Cloud, Apache Kafka, REST APIs, JWT, PostgreSQL, Docker, Docker Compose, API Gateway, Eureka

Tools & Platforms: Git, Twilio, WebSocket, SQLite, React, Streamlit, pgAdmin, Google Colab

CS Fundamentals: Data Structures & Algorithms, OOP, DBMS, Computer Networks, NLP, Vector Databases

WORK EXPERIENCE

Data Science & Computer Vision Intern — Drishti CPS Foundation, IIT Indore

July 2026

- Analyzed and validated manual and MediaPipe-derived facial morphometric measurements using statistical and regression techniques; performed data preprocessing, normalization, and visualization with Python, Pandas, NumPy, and Matplotlib.
- Evaluated prediction accuracy using Correlation, MAE, RMSE, Linear, Ridge, Lasso, and Deming Regression.

Research Intern — Machine Learning Security — SGSITS, Indore (Prof. Neha Mehra)

June 2026

- Researching adversarial machine learning and privacy-preserving techniques for secure AI systems using Negative Databases (NDBs); designing and evaluating NDB-based defense mechanisms against adversarial attacks while maintaining model accuracy.
- Implementing and benchmarking secure ML models using Python, NumPy, and Scikit-learn on standard datasets.

PROJECTS

CleanFlow — Microservices Booking Platform | [GitHub](#) ↗ | [Demo video](#) ↗

- Architected a production-grade 7-service Spring Boot system — auth-service (JWT + RBAC), booking-service with business rule engine, professional-service, notification-service, AI chatbot service, API Gateway, and Config/Discovery servers.
- Engineered Kafka async event pipeline (booking.events topic) for real-time cleaner alerts; containerized full infrastructure (PostgreSQL, Kafka, Zookeeper) with Docker Compose and deployed React frontend on Vercel.
- Centralized configuration via Spring Cloud Config Server with lb:// load-balanced Eureka routing for synchronous REST availability checks, plus full service-boundary documentation for production-style ops.

CEO Mind Clone — AI Executive Decision Chatbot | [GitHub](#) ↗

- Built Agentic RAG chatbot using LangGraph state machines delivering source-grounded executive decision support with full conversational memory across sessions.
- Engineered hybrid multi-stage retrieval (ChromaDB, BM25, RRF, cross-encoder reranking, MMR) with Self-RAG/CRAG self-correction loops for hallucination detection and response reliability.
- Implemented adaptive memory with autonomous re-querying and answer refinement when confidence thresholds are not met, improving precision and recall over naive single-retriever RAG.

AI Voice Receptionist | [GitHub](#) ↗ | [Demo video](#) ↗

- Built real-time AI voice system integrating Twilio Voice API and OpenAI Realtime API to fully automate inbound call handling with multi-turn dialogue and structured lead capture.
- Implemented bidirectional WebSocket audio bridge for low-latency G.711 μ -law streaming (sub-300ms round-trip); GPT-4o-mini post-call pipeline auto-generates summaries and persists leads to SQLite.
- Extracted structured lead attributes (name, intent, contact) from call summaries, enabling downstream CRM integration and analytics with zero manual intervention.

AI Research Automation System | [GitHub](#) ↗

- Engineered multi-agent research system using LangGraph orchestration and CrewAI agent roles spanning data ingestion, source validation, and publication-ready report generation.
- Designed modular graph-based workflow with specialized collection, summarization, and reasoning agents — enabling parallel execution and seamless extensibility for new research domains.
- Built distinct LLM-powered agents per pipeline stage — web/document ingestion, key-insight summarization, and analytical synthesis — without requiring pipeline redesign for new domains.